Nikhilesh Vijay Chirutala

Generative AI Developer

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SUMMARY

Generative AI Developer with a strong Python backend foundation, focused on building practical LLM-powered features for real-world applications. Experienced in developing RAG-based question-answering workflows, designing effective prompts, and integrating tools and APIs to connect AI capabilities with application data. Interested in creating context-aware, reliable AI experiences that support business workflows while maintaining clear response validation, sensible fallbacks, and a strong focus on usability and backend reliability.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Backend & Frameworks: FastAPI, Flask, REST APIs, API Development, Backend Automation

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, Jinja2, Microservices

AI & LLM: Generative AI Applications, Retrieval-Augmented Generation (RAG), LLM Integration, MCP Integration, Prompt Engineering

Databases: MongoDB, PostgreSQL, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman, API Integration

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging

WORK EXPERIENCE

Software Engineer • **CA One Tech Cloud Inc.** Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform *Python, FastAPI, AWS, Amazon S3, MySQL, LLM APIs, RAG, Datadog*

- Designed a GenAI-assisted call intelligence workflow to summarize transcripts and surface key topics for contact-center users through existing Python APIs.
- Prototyped retrieval over approved call and account context to generate grounded responses to agent questions and reduce manual searches.
- Implemented prompt templates and structured-output validation for call summaries, intent highlights, and follow-up suggestions, with checks for malformed responses.
- Integrated model inference into asynchronous Python service workflows, including request validation, timeout handling, retry policies, and persistence of generated results for downstream application use.
- Added dashboards and error reporting to track AI-service failures, latency, and processing outcomes for debugging and operational review.
- Designed human-review and fallback paths for uncertain or unavailable model responses, keeping generated content separate from source records and avoiding unsupported automatic actions.
- Evaluated representative prompts and sample inputs to check summary relevance, consistency with supplied context, and handling of empty or noisy transcripts.
- Worked across API, storage, and production-support layers to troubleshoot failures in the illustrative AI processing pipeline and maintain reliable integration with contact-center workflows.

Software Engineer • **Mantra Technologies** Oct 2023 – Mar 2026 | Hyderabad, India

- Built AI-assisted workflows connecting language-model responses with product data and business operations through Python services and REST APIs.
- Developed a RAG question-answering prototype over application records to generate context-aware responses from retrieved information.
- Created tool-calling interfaces for selected application operations, allowing an AI assistant to request approved reads or actions through explicit schemas and backend authorization checks.
- Designed prompt templates and response parsing for structured outputs, with validation and fallback behavior for missing fields, malformed results, and unsupported requests.
- Integrated AI features with frontend flows and backend APIs, including loading and error states, input validation, and persistence of relevant application results.
- Tested AI workflows against representative user questions and edge cases, reviewing relevance, groundedness, tool selection, and whether responses followed the requested output format.
- Implemented authentication and access-control checks around AI-connected application operations so model-generated requests could not bypass existing permissions.
- Investigated integration issues across model APIs, backend services, and application data, documenting limitations and improving reliability through defensive error handling.

Key Projects:

AI Bull — GenAI-Enabled Trading & Investment Platform *React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM*

- Developed an AI-powered trading and investment platform with portfolio aggregation, enabling users to consolidate holdings from multiple Demat accounts such as Upstox, Zerodha, and Groww into a unified view.
- Built portfolio workflows for tracking positions, stocks, and mutual funds, along with detailed reports and investment-level information across aggregated accounts.
- Developed stock and mutual fund screeners with configurable filtering capabilities and detailed asset pages for analyzing individual investments and their underlying information.
- Designed and implemented a backtesting engine for evaluating trading strategies against historical market data and analyzing strategy performance.
- Implemented a Retrieval-Augmented Generation (RAG)-based chatbot that combines portfolio and financial information as context to answer portfolio-related questions and provide contextual investment insights.
- Built Python and FastAPI APIs and business logic connecting portfolio aggregation, investment analysis, screening, and backtesting workflows with the platform's AI-enabled user experiences.

CRM.io — CRM Platform with AI-Assisted Workflows *FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM, MCP*

- Developed Python/FastAPI backend modules and REST APIs for CRM workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Built Contacts management with inline editing, global CRM search, related-record access, and Google Sheets integration for bulk contact imports.
- Developed a real-time CRM dialer integrated with Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts available to support relevant customer conversations.
- Implemented a CRM-wide soft-delete and Trash workflow with a 30-day retention period, replacing hard-delete behavior and using an AWS Lambda service to permanently remove records after the retention period.
- Built document management workflows using Amazon S3, including browser-to-S3 client uploads and storage of documents associated with CRM records.
- Developed a task cadence system that allows users to create sequences of tasks as a single workflow and progress through scheduled follow-ups, with Firebase Cloud Messaging (FCM) used for task notifications.
- Built account-level reporting dashboards and an MCP integration exposing CRM capabilities as tools for an AI bot to retrieve information and invoke operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground *Next.js, GitHub API, Vercel*

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform *Next.js, Neon DB, Clerk, Vercel*

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

EDUCATION

B.Tech — Electrical & Electronics Engineering 2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur